

Context-Length Order Recovery Study

A three-iteration investigation of whether context-length order leaves a persistent effect

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MODEL	4.8M-parameter character-level Transformer	COMPUT E	Apple M3, 24 GB

1. Question and original hypothesis

I asked whether the temporal order of context lengths used during training leaves a persistent effect on final language-model performance. This is a training-dynamics question, not a search for the best hyperparameter: each schedule used the same context lengths and training budget, but presented the context blocks in a different order.

My original interpretation was that descending training (256-128-64-32) created a persistent disadvantage relative to ascending training (32-64-128-256). If that interpretation were correct, a substantial descending-minus-ascending gap should survive after both models received the same final long-context training condition.

Research progression

Iteration	Design change	What it allowed me to learn
1. Exploratory	One seed; schedule and diagnostic probes	A large deficit appeared, but measurement validity was weak.
2. Paired	Five paired seeds; shared initialization and data manifests; fixed held-out targets	The terminal deficit replicated, but recent context was still confounded with order.
3. Recovery	Common T=256 recovery; four evaluation horizons; process and anchor panels; preregistered rules	The large deficit did not persist; the smaller reversed residual remained unresolved.

2. Why the original experiment could not answer the question

Iteration 1 used one seed, unpaired batches, inconsistently aligned evaluation targets across context horizons, and a corpus probe affected by resubstitution bias. Its dramatic result was therefore a reason to investigate, not a causal conclusion.

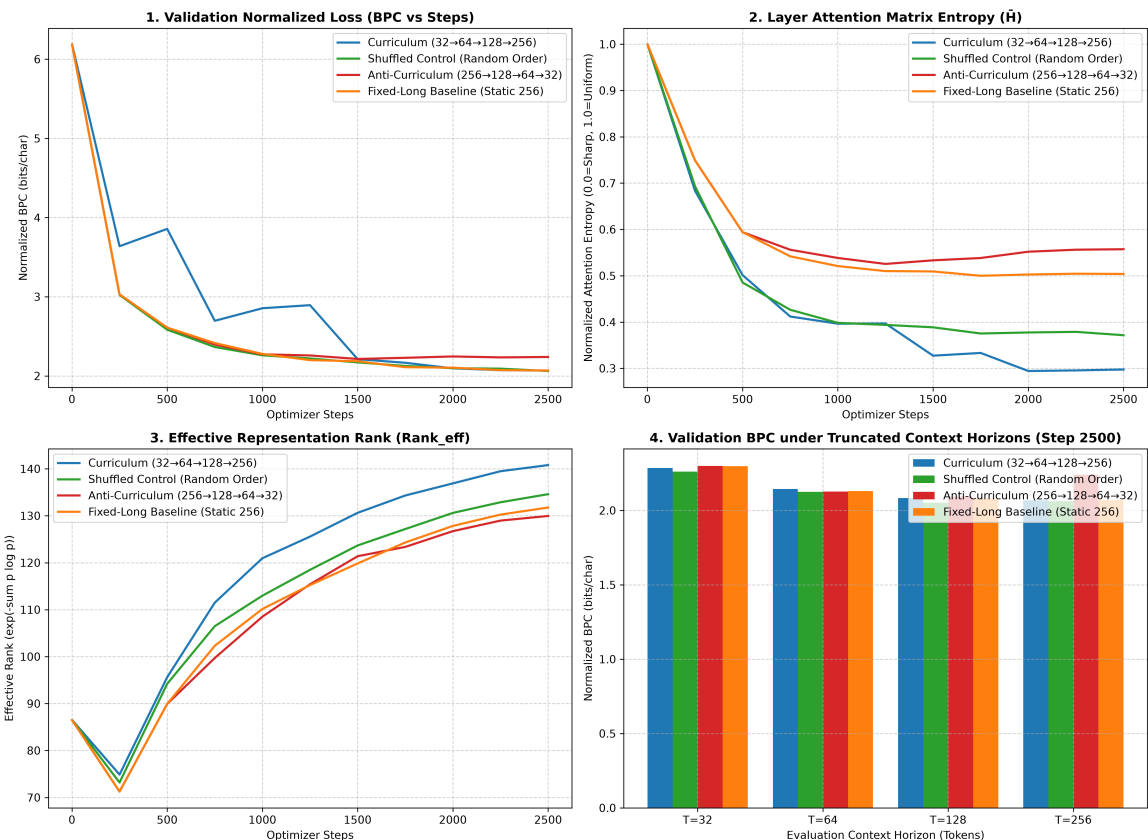


Figure 1 | Iteration 1: an informative but unreliable starting point. The pilot suggested schedule-dependent differences in loss and diagnostics. Several mechanism claims were later weakened or retracted because the measurement and pairing design could not support them.

Iteration 2: correcting measurement and reproducibility

I introduced five paired seeds, identical initialization within each seed, context-specific shared data manifests, held-out validation text, fixed target positions, and corrected same-target evaluation. The descending terminal deficit remained consistent, so the observation was no longer a one-seed accident.

However, the stronger design exposed a deeper confound: ascending training ended at $T=256$, descending training ended at $T=32$, and the principal evaluation also used $T=256$. The result could reflect persistent ordering, recent-context adaptation, or an interaction between context length, optimizer history, and the late learning-rate phase.



Figure 2 | Iteration 2: the terminal deficit replicated under paired measurement. Five paired seeds made the endpoint contrast more credible, but did not identify its cause. The figure is retained as historical evidence; its original mechanism language was stronger than the design justified.

The key methodological lesson was that more seeds cannot repair a confounded estimand. Iteration 2 estimated the terminal contrast more precisely, but a new intervention was needed to distinguish persistence from recency.

3. How the recovery study controlled the comparison

Iteration 3 gave ascending, descending, and one prespecified nonmonotonic schedule four 500-step context blocks, followed by 500 identical T=256 recovery steps. The confirmatory comparison remained descending versus ascending; the nonmonotonic schedule was exploratory.

Within each of three paired seeds, I held constant model architecture, initial weights, tokenizer, corpus split, exposure to every context length, 4,096 target tokens per update, context-specific data draws, recovery draws at every step, AdamW settings, preserved optimizer state, global learning-rate schedule, training budget, validation sequences, target positions, and scoring code. Only the temporal permutation of the four pre-recovery blocks varied.

A fixed 16-sequence process panel measured all four horizons (T=32, 64, 128, 256) at 17 checkpoints. A nested 32-sequence anchor panel supplied the formal endpoints. Estimands, thresholds, checkpoints, figures, and decision rules were frozen before the formal results were inspected.

4. Core result: +0.7570 to -0.0485 BPC

Before recovery, the paired descending-minus-ascending anchor gap at T=256 was **+0.7570 BPC** (95% paired t interval [+0.3290, +1.1851]; 3/3 seeds positive). The large deficit therefore replicated under the matched-block design.

After 500 common T=256 steps, the same contrast was **-0.0485 BPC** (95% interval [-0.1491, +0.0521]; 0/3 positive). On the process panel, the gap fell from +0.8097 at recovery step 0 to +0.0719 after 10 steps and crossed zero by 50 steps.

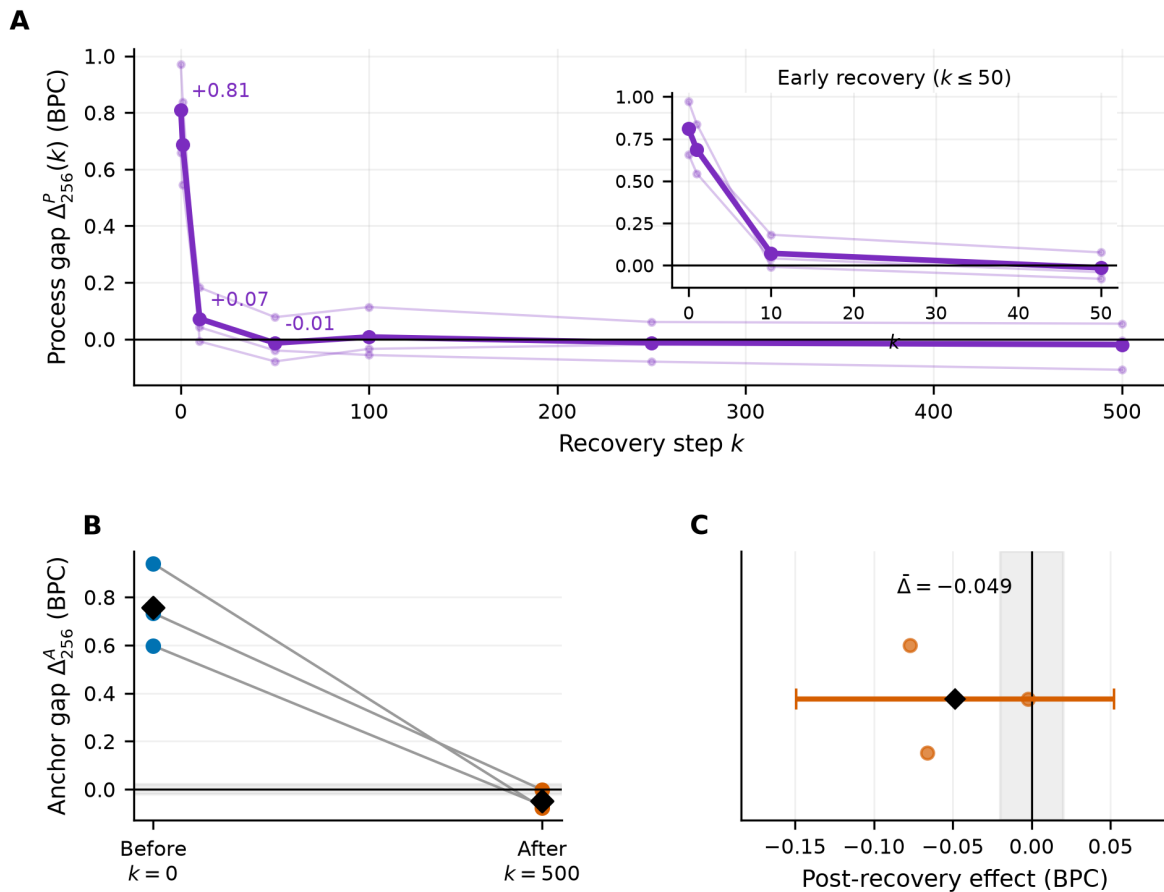


Figure 3 | Iteration 3: common long-context recovery rapidly reversed the large deficit. Panel A shows the recovery trajectory, Panel B the three paired seed endpoints, and Panel C the uncertain post-recovery mean. The registered endpoint lies outside the ± 0.02 BPC practical-removal band, so the smaller sign reversal remains unresolved.

5. Why the original interpretation was weakened

A large persistent ordering effect should have remained substantially positive after the arms entered the same long-context condition. Instead, most of the gap disappeared within the first few recovery checkpoints. Before recovery, the large contrast was also specific to T=256 evaluation; the corresponding contrasts at T=32, 64, and 128 were already small and negative.

The reversal was driven mainly by improvement in the descending arm. Its mean T=256 BPC improved from 2.9980 to 2.2345, while the ascending mean changed only from 2.2410 to 2.2830. The result was therefore not created simply by deterioration of the reference arm.

This pattern is more consistent with recent-context adaptation than with a large persistent ordering effect. It does not uniquely prove a recency mechanism: recovery added both long-context exposure and 500 optimizer updates, and may have acted through parameter adaptation, optimizer state, representation changes, or learning-rate interactions.

The experiment rejects the large persistent-deficit interpretation more strongly than it identifies a unique replacement mechanism.

6. What remains unknown

The immediate unresolved question is whether the approximately -0.05 BPC post-recovery contrast replicates. With three paired seeds, its confidence interval includes both a moderate negative effect and a small positive effect. The current study cannot claim equivalence or establish that the sign reversal is a real smaller ordering effect.

The study also does not determine:

- which component of recovery caused the rapid reversal;
- whether resetting optimizer state at transitions would change the trajectory;
- whether the finding generalizes beyond this model, tokenizer, corpus, optimizer, context range, and token budget;
- whether other monotonic or nonmonotonic schedules behave similarly;
- whether longer matched recovery would remove, preserve, or amplify the small negative residual.

Defensible conclusion

The large deficit associated with descending context order did not persist under matched T=256 recovery. Its rapid reversal is more consistent with a recent-context effect than with persistent ordering, although a smaller ordering effect cannot yet be ruled out.

Code, preregistration, complete technical report, and formal artifacts: github.com/angelazu-builder/nanoGPT